ETSI TS 126 450 V17.0.0 (2022-05)



Universal Mobile Telecommunications System (UMTS);
LTE;
Codec for Enhanced Voice Services (EVS);
Discontinuous Transmission (DTX)
(3GPP TS 26.450 version 17.0.0 Release 17)



Reference
RTS/TSGS-0426450vh00

Keywords
LTE,UMTS

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from:
http://www.etsi.org/standards-search

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format at www.etsi.org/deliver.

Users of the present document should be aware that the document may be subject to revision or change of status.
Information on the current status of this and other ETSI documents is available at
https://portal.etsi.org/TB/ETSIDeliverableStatus.aspx

If you find errors in the present document, please send your comment to one of the following services:
https://portal.etsi.org/People/CommiteeSupportStaff.aspx

If you find a security vulnerability in the present document, please report it through our
Coordinated Vulnerability Disclosure Program:
https://www.etsi.org/standards/coordinated-vulnerability-disclosure

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm except as authorized by written permission of ETSI.
The content of the PDF version shall not be modified without the written authorization of ETSI.
The copyright and the foregoing restriction extend to reproduction in all media.

© ETSI 2022.
All rights reserved.

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI Web server (https://ipr.etsi.org/).

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™** and **LTE™** are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Legal Notice

This Technical Specification (TS) has been produced by ETSI 3rd Generation Partnership Project (3GPP).

The present document may refer to technical specifications or reports using their 3GPP identities. These shall be interpreted as being references to the corresponding ETSI deliverables.

The cross reference between 3GPP and ETSI identities can be found under http://webapp.etsi.org/key/queryform.asp.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

Contents

Intellectual Property Rights 2
Legal Notice 2
Modal verbs terminology 2
Foreword 4
1 Scope 5
2 References 5
3 Abbreviations 6
4 General 6
4.1 Background 6
4.2 General organization 6
5 EVS Codec DTX Operation 7
5.1 Transmit (TX) side 7
5.2 Receive (RX) side 7
Annex A (informative): Change history 8
History 9

Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

This document specifies the system level aspects of the Discontinuous Transmission (DTX) function of the EVS codec.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 26.445: "Codec for Enhanced Voice Services (EVS); Detailed Algorithmic Description ".

[3] 3GPP TS 26.442: " Codec for Enhanced Voice Services (EVS); ANSI C code (fixed-point)".

[4] 3GPP TS 26.443: " Codec for Enhanced Voice Services (EVS); ANSI C code (floating-point)".

[5] 3GPP TS 26.444: " Codec for Enhanced Voice Services (EVS); Test Sequences".

[6] 3GPP TS 26.446: " Codec for Enhanced Voice Services (EVS); AMR-WB Backward Compatible Functions".

[7] 3GPP TS 26.447: " Codec for Enhanced Voice Services (EVS); Error Concealment of Lost Packets".

[8] 3GPP TS 26.448: " Codec for Enhanced Voice Services (EVS); Jitter Buffer Management".

[9] 3GPP TS 26.449: " Codec for Enhanced Voice Services (EVS); Comfort Noise Generation (CNG) Aspects".

[10] 3GPP TS 26.441: " Codec for Enhanced Voice Services (EVS); General Overview".

[11] 3GPP TS 26.451: " Codec for Enhanced Voice Services (EVS); Voice Activity Detection (VAD)".

[12] 3GPP TS 26.114: "Multimedia Telephony; Media handling and interaction".

[13] 3GPP TS 26.131: "Terminal acoustic characteristics for telephony; Requirements ".

[14] 3GPP TR 26.952: "Codec for Enhanced Voice Services (EVS); Performance Characterization".

[15] 3GPP TS 26.452: "Codec for Enhanced Voice Services (EVS); ANSI C code; Alternative fixed-point using updated basic operators".

3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

ACELP	Algebraic Code-Excited Linear Prediction
AMR-WB	Adaptive Multi Rate Wideband (codec)
CNG	Comfort Noise Generator
DTX	Discontinuous Transmission
EVS	Enhanced Voice Services
FB	Fullband
FEC	Frame Erasure Concealment
IP	Internet Protocol
JBM	Jitter Buffer Management
MSB	Most Significant Bit
MTSI	Multimedia Telephony Service for IMS
NB	Narrowband
PS	Packet Switched
PSTN	Public Switched Telephone Network
SAD	Signal Activity Detection
SC-VBR	Source Controlled - Variable Bit Rate
SID	Silence Insertion Descriptor
SWB	Super Wideband
VAD	Voice Activity Detection
WB	Wideband
WMOPS	Weighted Millions of Operations Per Second

4 General

4.1 Background

The present document specifies the system level aspects of the Discontinuous Transmission (DTX) function of the EVS codec. Specific codec functions that implement the DTX and other EVS functionalities are described in TS 26.445 [2], TS 26.446 [6], TS 26.447 [7], TS 26.448 [8], TS 26.449 [9] and TS 26.451 [11]. The corresponding fixed-point reference C code is specified in TS 26.442 [3] and TS 26.452 [15] using two different sets of basic operators. The corresponding floating-point reference C code is specified in TS 26.443 [4]. The EVS test sequences are given in TS 26.444 [5] and the EVS codec is characterized in TR 26.952 [14].

Discontinuous transmission (DTX) is a mechanism for the EVS Codec to encode the input signal at a lower average rate by taking speech inactivity into account. The DTX scheme is useful for the following purposes:

- to conserve UE power,
- to improve network capacity.

DTX in the transmitting path (uplink) shall be in operation in UEs if commanded so by the network. The UE shall handle DTX in the receiving path (downlink) at any time, regardless of whether DTX in the transmitting path is commanded or not.

4.2 General organization

The default DTX mechanism described in the present document requires the following functions:

- a Signal Activity Detector (SAD) on the transmit (TX) side;
- evaluation of the background acoustic noise on the transmit (TX) side in order to transmit characteristic parameters to the receive (RX) side;

- generation, on the receive (RX) side, of a similar noise called comfort noise during periods where the transmission is switched off.

The Signal Activity Detector (SAD) is defined in TS 26.451 [11] and the comfort noise functions in TS 26.449 [9]. Both are based partly on the EVS Codec and its internal variables defined in TS 26.445 [2].

In addition to these functions, if the parameters arriving at the RX side are detected to be seriously corrupted by errors, the speech or comfort noise shall be generated from substituted data in order to avoid seriously annoying effects for the listener. These functions are defined in TS 26.447 [7].

An overall description of the speech processing parts can be found in TS 26.441 [10]. An overview of the DTX operation is shown in Figure 1.

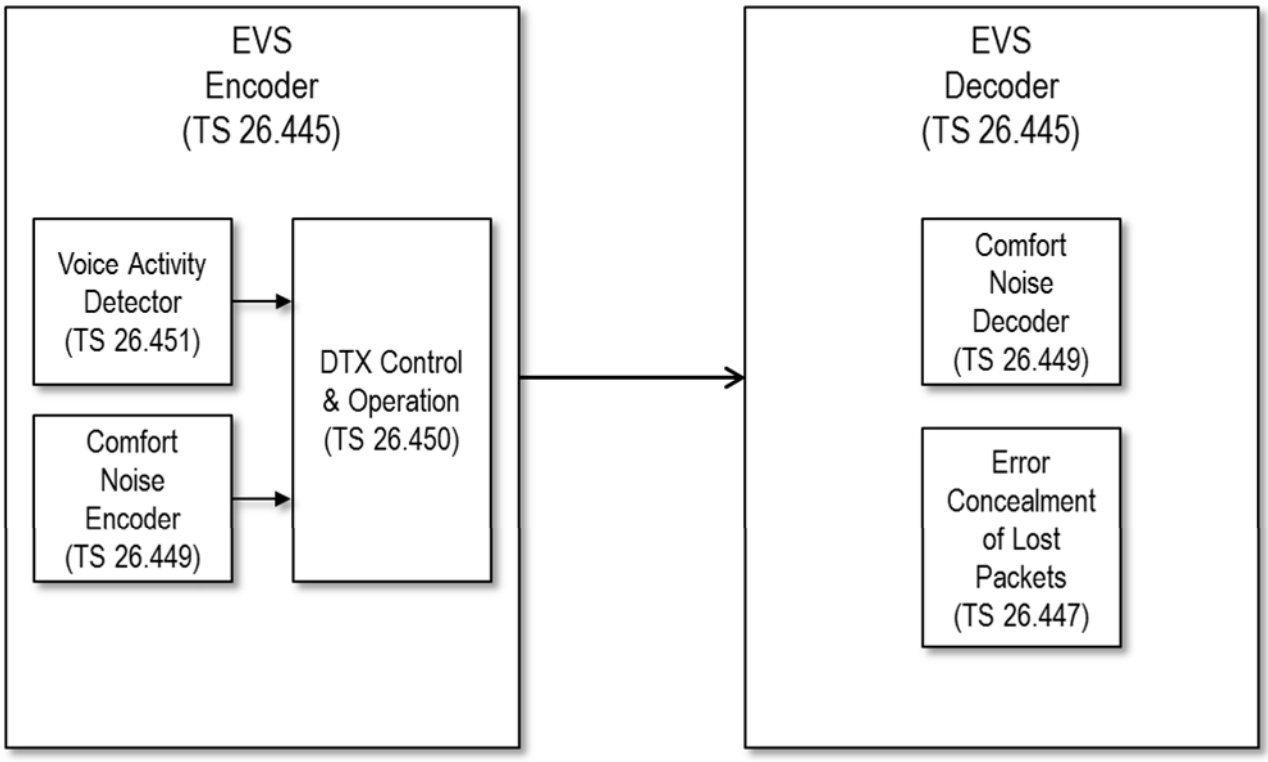


Figure 1: Block diagram of DTX Operation

5 EVS Codec DTX Operation

5.1 Transmit (TX) side

There are two algorithms for extending the DTX hangover period. One is a general hangover algorithm and other is specific to music. Both are defined in clause 5.1.12.5 of TS 26.445 [2].

The main specification of the DTX/CNG algorithms is contained in clause 5.6 of TS 26.445 [2].

The bit allocation of SID frames generated for DTX operation is defined in clause 7.2 of TS 26.445 [2].

5.2 Receive (RX) side

Operation of the EVS decoder, including decoding of SID frames generated during DTX operation is described in clause 6.7 of TS 26.445 [2].

Annex A (informative): Change history

Change history							
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**
2014-09	65	SP-140465			Presented at TSG#65 for approval		1.0.0
2014-09	65				Approved at TSG SA#65	1.0.0	12.0.0
2015-12	70				Version for Release 13	12.0.0	13.0.0

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2017-03	75					Version for Release 14	14.0.0
2018-06	80					Version for Release 15	15.0.0
2019-03	83	SP-190036	0001	-	B	Addition of reference to Alt_FX_EVS implementation	16.0.0
2022-04	-	-	-	-	-	Update to Rel-17 version (MCC)	**17.0.0**

History

Document history		
V17.0.0	May 2022	Publication